13-xy-windows: joint x and y window factor bases vs x-only windows of matched size

Pair-46 factor-base campaign, item 13, run 2026-10-15. The protocol was frozen first in `PROTOCOL.md` (sha256 `f7224755...f876` in `PROTOCOL.sha256`). The run had one execution and no dry run. An earlier launch failed before writing any data because `/usr/bin/time` was missing.

Host: shared 4-CPU container, run pinned to 1 core (`taskset -c 3`). Wall time was 364.9 s: 31 s for pair 46 and 334 s for the toy curves. The budget guard did not fire, so nothing was cut.

Verdict: null, with an exact obstruction for the sign and parity forms.

• **Coverage.** The existing `bit_window` family does not cover joint windows. It is x-only and unimplemented, so this item implemented joint windows.

• **Sign and parity windows (exact).** A y-sign window (y < p/2) or a y-parity window adds nothing: its x-set is exactly the x-only window. This was checked on 200/200 curves and 8000/8000 targets.

• **Box and bit windows (empirical).** At matched size, |y|-box windows and y-bit windows give the same relation yield as the x-only window. The decompositions-per-target ratio D/D(X) is 0.9885 [0.9608, 1.0181] for box 1/4, 0.9857 [0.9562, 1.0168] for box 1/16, 0.9921 [0.9629, 1.0216] for box 1/64 and 0.9946 [0.9631, 1.0257] for y-bits 1/16.

• **Cost.** These windows cost 2^k times more x-candidates to build: 8.00, 31.9, 127.8 and 32.0 scans per element, against 2.00 for the x-only window.

• **Controls.** Both positive controls fired: PC M2 ratio 0.0004 and X_half D ratio 0.254.

• **Pair 46.** Box counts in (x, |y|) are as random as on 4 random curves over the same p.

Step 0: is this already covered by bit_window?

No.

• `research/factor_base_lab.md` (crypto-autoresearcher) defines `bit_window` as "fixed high or low radix bits of **x**".

• `src/crypto_autoresearcher/index_calculus/factor_base.py` implements small_x, random, subgroup, dickson, known_log, planted and small_x_offset, and none of them reads y. Neither `bit_window` nor `qr_window` is implemented.

• The ecbench plug-in families in `docs/ecbench/FACTOR_BASE_PROTOCOL.md` are x-only.

• The ledger proposal IDEA-20261015-5888a7 ("joint x-y windows", notes: "nothing measured") records the same gap.

This item is the first measurement. Its x-only comparator `Xbits4` is the bit_window family, instantiated as x = 0 mod 16.

Exact statements (proved; computation only checks them)

• **X1. Every joint window is an x-predicate.**

• Statement: for Y a subset of F_p, take the base {P : x(P) in X, y(P) in Y} modulo +-. Its x-set is {x in X : f(x) is a nonzero square and {+-sqrt f(x)} meets Y}.

• Proof: Semaev relations R = sum +-P_i and S_{m+1}(x_1..x_m, x(R)) = 0 depend only on x(P_i). So a y-constraint only selects which x enter the base. It is a thinning of the x-line by the predicate "a root of y^2 = f(x) lies in Y", and it adds no algebraic relation.

• **X2. Sign and parity windows are no-ops.**

• Statement: if Y and -Y together cover F_p^*, the x-set equals that of the x-only window, and every

relation is identical, target by target.

• The half-plane y < p/2 is such a Y. So is "y even": p is odd, so y -> p - y flips parity, and exactly one of +-y is even.

• Check: the XY_sign and XY_par sets equalled X, and their per-target decomposition counts were identical, on 200/200 curves and 8000/8000 targets.

• Consequence: the "y sign/bit constraint" contributes nothing unless it is a genuine thinning of |y| = min(y, p - y).

• **X3. Membership degree is not lowered.**

• A base of K distinct abscissae has univariate vanishing polynomial degree exactly K. A nonzero function on E vanishing at K affine points has pole order >= K, since deg div = 0.

• A y-window therefore cannot give a membership polynomial of degree far below the base size. This is the campaign's standing prime-field obstruction (items 06 and 10, RR), specialised here.

• Imposing |y| < B2 on a Semaev solve also needs extra variables y_i with y_i^2 = f(x_i) and a smallness condition on y_i. That condition is no more algebraic than small-x is.

• **X4. Price.**

• A box of fraction q of the |y| range keeps q of the liftable x. Building K elements by x-scan therefore costs 2/q candidates per element, against 2 for the x-only window.

• This is the expected value under equidistribution of |y|. Matched-size yield cannot repay it, because the yield is conserved (below).

• A box enumerator cheaper than the x-scan (c_box < B1 in IDEA-20261015-5888a7) is not known. None is implemented here; the open number stays open.

Empirical measurements: toy relation yield at matched size

• Curves: 200 toy curves, 20 to 30 bits. There are 100 family curves (F: p = 5t^2+19t+19, j = -96^3) and 100 random prime-order curves (N2).

• Size: every arm holds exactly K = ceil(sqrt(N/2)) +- classes; X_half holds K/2.

• Targets: T = 40 random targets per curve, shared by all arms.

• Relations: Semaev m = 2, R = +-P1 +- P2.

• Ratios: pooled sum over curves of arm / X, with a 95% bootstrap over curves (2000 resamples).

• Baseline: X has M2 = 0.6349 [0.6243, 0.6454] (Wilson) and D = 1.014 decompositions per target.

arm	predicate	M2 / M2(X) [95%]	D / D(X) [95%]	x scanned per element [95%]
X	first K liftable x (x-only window)	1	1	1.9995 [1.9959, 2.0031]
XY_sign	some root y < p/2	1.0000 (identical, X2)	1.0000 (identical)	1.9995
XY_par	some root y even	1.0000 (identical, X2)	1.0000 (identical)	1.9995
XY_box2	abs y < p/8	0.9949 [0.9732, 1.0166]	0.9885 [0.9608, 1.0181]	7.9969 [7.9796, 8.0149]
XY_box4	abs y < p/32	0.9813 [0.9594, 1.0030]	0.9857 [0.9562, 1.0168]	31.943 [31.861, 32.026]
XY_box6	abs y < p/128	0.9992 [0.9770, 1.0235]	0.9921 [0.9629, 1.0216]	127.76 [127.44, 128.07]
XY_bits4	bits 1..4 of abs y zero	0.9927 [0.9708, 1.0156]	0.9946 [0.9631, 1.0257]	32.016 [31.934, 32.100]
Xbits4 (x-only bit_window)	x = 0 mod 16	0.9941 [0.9717, 1.0163]	0.9905 [0.9605, 1.0198]	32.046 [31.987, 32.106]
R_random (neg. control)	K random liftable x	1.0012 [0.9784, 1.0259]	0.9860 [0.9556, 1.0173]	2.0010
PC_logAP (pos. control)	{jG : j = 1..K}	0.0004 [0.0000, 0.0010]	0.414 [0.000, 1.183]	1
X_half (size control)	first K/2 of X	0.3585 [0.3442, 0.3722]	0.2538 [0.2443, 0.2630]	1.0006

Notes on the table:

• "abs y" is the canonical |y| = min(y, p - y). A box fraction 2^-k means abs y < ((p+1)/2)/2^k.

• PC_logAP's D is dominated by 2 targets whose log fell in [-2K, 2K]. Those targets carried thousands of

decompositions; every other target had none. That is why its D interval is wide while M2 collapses.

Both the F and the N2 subgroup give the same picture; every joint, Xbits4 and R interval is inside [0.92, 1.07] (`results.json`, groups F and N2).

Against the preregistered decision rule:

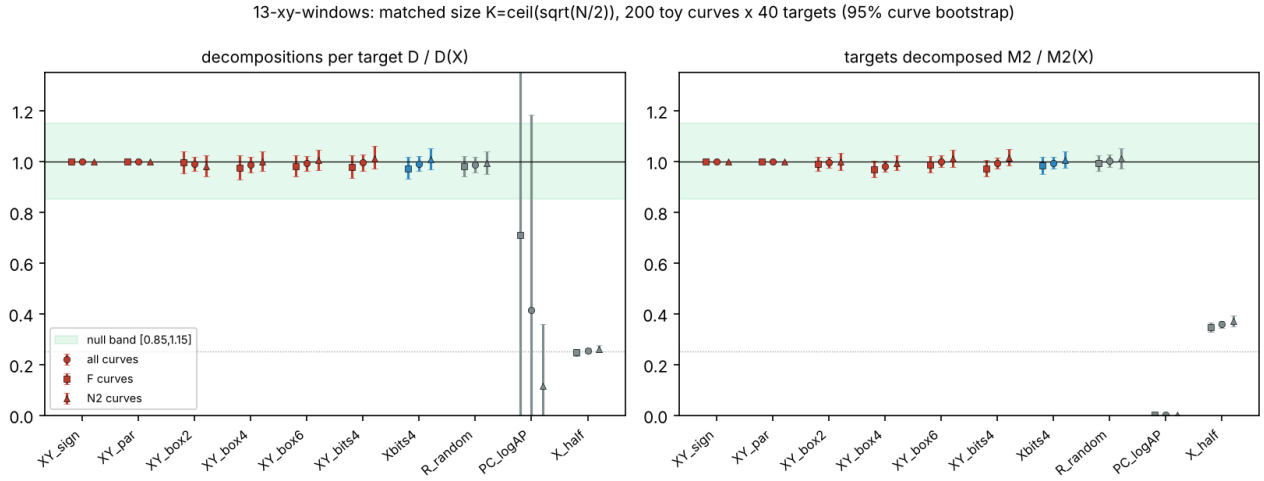
• Every joint arm and Xbits4 has both intervals inside [0.85, 1.15]. In fact all lie inside [0.95, 1.03].

• R is inside the band.

• PC M2 is < 0.5 and its interval excludes 1.

• The X_half D interval contains 0.25.

Null: yield is conserved at matched size. A y-window thins the x-line like any other predicate, and the base behaves like a random set of the same size. The effect is the same as for the x-only bit_window. Two joint arms are at the edge of the band, with upper bounds just above 1.0: XY_box4 at 1.003 for M2, and the F subgroup at 1.002. No arm shows a gain.



Matched-size yield ratios against the x-only window (95% curve bootstrap)

Empirical measurements: pair 46 box-count statistics

Setup:

• For crater, floor and 4 random curves over the same p, the run scanned x in [0, 2^23).

• 2^22 x were liftable on each (e.g. crater 4,194,446). f(x) = 0 never occurred.

• For crater and floor, N·P = O was checked on 3 random points per curve.

Dispersion of box counts. Each box is [x0, x0 + B1) x [c, c + H) in (x, |y|), with lambda ~ 16.

B1	crater	floor	random0..3 range	planted crater (+8 in 10% of boxes)	exact null 1 - q (post hoc)
2^8	0.9375 [0.9324, 0.9427]	0.9364 [0.9312, 0.9417]	0.9354 .. 0.9399	1.2316 [1.2249, 1.2386]	0.9375
2^12	0.9973 [0.9920, 1.0026]	0.9924 [0.9869, 0.9980]	0.9934 .. 0.9989	1.2856 [1.2784, 1.2933]	0.99609
2^16	1.0001 [0.9945, 1.0054]	0.9959 [0.9909, 1.0010]	0.9943 .. 1.0037	1.2935 [1.2865, 1.3007]	0.99976

Deviation from the protocol, reported rather than hidden. The preregistered null was I = 1, which is Poisson. That is wrong for this instrument:

• Each x contributes at most one point (one canonical |y|), so a box count is Binomial(B1, q) with q = lambda/B1.

• The exact null is therefore var/mean = 1 - q: 0.9375, 0.99609 and 0.99976.

Effect on the preregistered rule:

• Read literally, the rule fails at B1 = 2^8 for **all six curves, random ones included**. That is a misspecified null, not structure.

• Against the exact null (computed post hoc in `posthoc.py`/`posthoc.json` from the recorded values), all 18 intervals contain it. Index/null ratios run from 0.9946 to 1.0040.

• The planted control passed at every shape: the interval excludes 1 and lies above 1.2. At B1 = 2^8 the margin is small (1.2316, from a 0.9375 baseline).

Pair-46 curves are inside the range of the random curves at every shape.

Corner density. Within x in [0, 2^23), this is the fraction of liftable x whose |y| < ((p+1)/2)/2^k, for k = 0..16, as observed/expected.

• All six curves track 1. At k = 16 there are about 64 points, so the spread is ±25%.

• Intervals excluding 1 occur on:

• floor, k = 4..9 and k = 11, at 0.95 to 0.995.

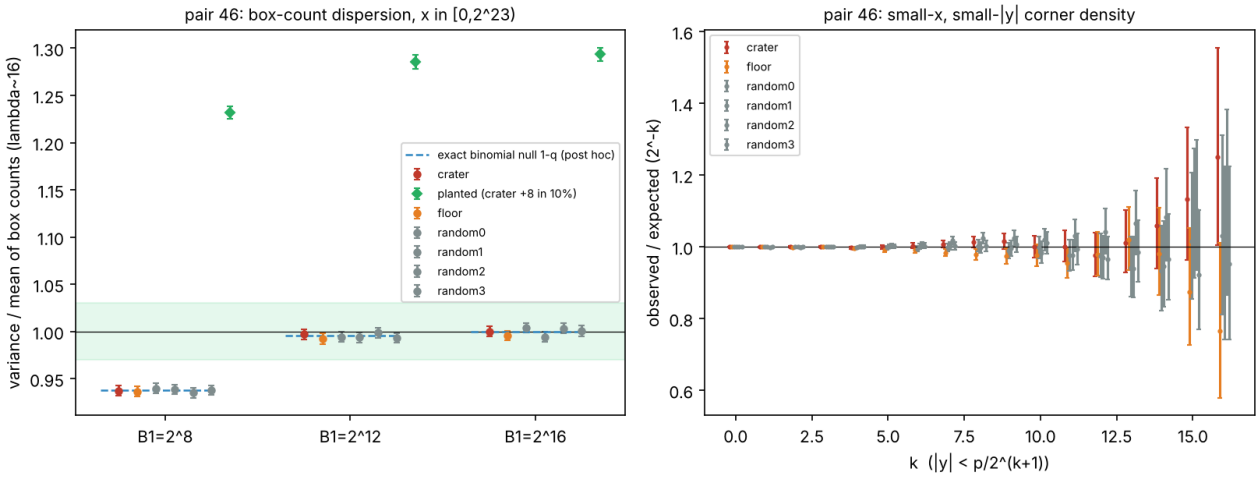
• random2, k = 1, 2 and 7..9, at 1.02 to 1.03.

• crater, k = 16 (1.25).

• The k-levels are nested cumulative counts, so these exclusions are one correlated excursion per curve, not independent findings.

• A random curve shows an excursion of the same size in the opposite direction. There is no family-specific signal, and these are not corrected for the 6 x 16 comparisons.

The realized pair-46 density of the XY_box_k base is 2^-k of the liftable x. Its x-scan price is therefore 2^(k+1) candidates per element at pair-46 size too.



Pair 46: box-count dispersion vs the exact binomial null, and the small-x/small-|y| corner density

What is exact and what is empirical

Exact:

• X1: a joint window is an x-predicate.

• X2: sign and parity windows are identical to the x-only window (checked 8000/8000).

• X3: membership degree >= K.

• X4: the scan price is 2/q per element, given equidistribution. The measured 2^(k+1) scans per element confirm it.

Empirical:

• Matched-size yield conservation for |y| boxes and y-bit windows (toy, 20 to 30 bits, m = 2).

• Pair-46 equidistribution of (x, |y|) box counts in x < 2^23, at box scales far below the Weil range: B1·H ~ 1e17, against p^{3/2} ~ 4.6e23.

Not measured:

• m >= 3 relations.

• A cheaper-than-x-scan box enumerator (c_box). No candidate exists to test.

Files and reproduce

Files:

• Frozen protocol: `PROTOCOL.md`, `PROTOCOL.sha256`.

• Instrument: `src/main.rs` (Rust, single thread) and `src/common.rs` (copied from item 05).

• Input curves: `input_curves_from_04.csv` (sha256 88c26329...7c12).

• Raw outputs: `results.json` (all recorded values, per-curve rows), `run.log`, `stdout.txt`, `time.txt`.

• Post-hoc exact-null comparison: `posthoc.py` → `posthoc.json`.

• Figures: `plot.py` → `fig_yield.{svg,png}`, `fig_pair46.{svg,png}`.

• PDF layout: `make_pdf.py` → `report.pdf`.

```
cd research/pair46_factor_bases_20261015/13-xy-windows
CARGO_TARGET_DIR=/tmp/t13 cargo build --release
taskset -c 3 /tmp/t13/release/xy_windows . 40 100 100 2000 1500   # out_dir T nF nN2 boots toy_budget_s
python3 posthoc.py . && python3 plot.py . && python3 make_pdf.py .
```